



An empirical evaluation of clustering processes for early detection of university dropout

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Abstract

The elevated rates of dropout within academic institutions have prompted the use of Artificial Intelligence (AI) to tackle this issue. These efforts often rely mainly on administrative and academic data, lacking personal information about students. In a previous study, we explored machine learning models to leverage this data and harness their knowledge-extraction capabilities. However, a critical factor, the availability of labeled data, was not addressed. Obtaining these data may be challenging due to their distribution across different systems or the considerable time required to collect them, especially when new degrees are being implemented. The lack of labeled data is a major obstacle for institutions that do not possess them so that they are unable to take advantage of the full potential of AI for their purposes. Clustering algorithms have conventionally been employed to uncover latent patterns within unlabeled data. These unsupervised algorithms may reduce the need for data labeling; nonetheless, it necessitates rigorous validation of the resulting clusters, particularly when dealing with datasets encompassing numerical and categorical attributes. This paper introduces a comparison of various clustering algorithms to discern the most appropriate technique for uncovering the underlying factors contributing to university student attrition, employing unlabeled data. The novelty lies not only in the algorithmic comparison but also in their integration with diverse data preprocessing methodologies, streamlining the selection of the optimal combination including advanced data transformations for the harmonization of numerical and categorical information. It is illustrated through a real-world case utilizing academic data from a Spanish university, providing empirical validation for the proposed methodology. We also conducted an exploratory analysis to identify the factors behind cluster formation. The insights gained can be extrapolated to analogous experiments where social or economic data are scarce, and most of the available attributes are academic in nature.

Keywords Machine learning · Feature engineering · Not supervised models · Clustering models · Real experiences · Student dropout

1 Introduction

In the last years, Spain has been among the countries with the least efficient utilization of public and private higher education due to elevated dropout rates (over 30% [1]). Being academic institutions aware of this issue, they have been exploring the potential offered by Artificial Intelligence to create predictive models that could foresee, prevent, and if needed, establish actions to lessen dropout rates. These predictive models that anticipate student attrition may be built by using different approaches, depending on the nature of the data used for their creation. Some of these approaches utilize personal information about the students and families (such as social or economic aspects); others leverage interaction data gathered by online learning platforms to analyze

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student behavior and engagement with educational content; finally, there are scenarios where institutions only have access to academic data without significant personal details, that is, strictly necessary administrative and academic information from the student.

In our previous work [2, 3], we focused on predicting student dropout using only academic data, with minimal socio-economic or demographic information. To enable early detection, we developed models tailored to different stages of the student lifecycle, from pre-enrollment to later academic progress. Given the limited availability of comprehensive data, the study emphasized maximizing the predictive value of existing academic records through intensive preprocessing, feature engineering, and rigorous validation. Nevertheless, despite making significant contributions to detecting academic dropout, it did not address a key limitation in dropout prediction: the lack of sufficient labeled data. Acquiring labeled data in academic contexts is particularly challenging, as it often requires tracking student trajectories over several years. The process is not only time-consuming but also prone to inconsistencies due to factors such as changes in academic programs, curricular reforms, institutional transfers, temporary withdrawals, or joint degree structures.

In light of the current context, there arises a crucial research question that would be very relevant to address: whether it is possible to develop an unsupervised model capable of detecting academic dropout among students, even with data that have not been previously labeled. Exploring the inherent structure of the data can enable predictive capabilities without relying on pre-assigned labels. This is especially relevant given that most studies on academic dropout highlight the importance of early risk identification. Our work addresses this need by supporting the deployment of predictive models even in the absence of labeled data, thus providing timely and actionable insights for educational institutions.

Clustering algorithms are widely used in the literature to uncover hidden patterns in unlabeled data. Applying unsupervised learning to detect dropout reduces labeling costs and facilitates the deployment of predictive models across institutions. However, it poses validation challenges through the clusters created and requires thorough evaluation. In this context, a key question arises: Can two distinct clusters be effectively created without using the dropout feature and subsequently validated against labeled data to assess their separation accuracy?

While many studies have applied clustering techniques to numerical attributes, fewer have explored their effectiveness with categorical data, which are prevalent in academic datasets. Real-world data typically combine numerical and categorical variables, each offering essential insights into complex patterns. Therefore, successful analysis requires

careful preprocessing and harmonization of these variable types. In this context, data transformations play a critical role in improving the performance of unsupervised algorithms. Moreover, clustering results are often not validated against labeled data, making interpretation more difficult, especially when clustering is based on categorical features and involves mixed-type variables. Although some studies address these scenarios, optimization of clustering algorithms for mixed data remains underexplored. Enhancing such optimization, particularly through appropriate preprocessing and feature engineering, can be key to identifying the most suitable algorithms for academic dropout detection.

Our research makes significant contributions in three key aspects:

- We present an approach for comparing multiple unsupervised algorithms to identify dropout-related patterns, using the dropout label only for evaluation. The method aims to form two clusters that effectively separate students without relying on labeled data. Moreover, as the study relies primarily on academic records, it offers high replicability across institutional contexts.
- Our study applies advanced data transformations to effectively preprocess and harmonize numerical and categorical features, enhancing the robustness of unsupervised algorithms through a tailored treatment of their distinct properties.
- We validate the proposed approach using both clustering metrics and labeled data, going beyond standard unsupervised evaluation. This dual validation confirms the reliability and generalizability of our method, demonstrating its practical value through application to a real-world dataset that reflects the complexity of actual academic scenarios and showcases the tangible benefits it brings to practical applications.

The remainder of the paper is structured as follows. Section 2 reviews relevant literature on student dropout and the use of unsupervised techniques, including a comparative analysis of prior studies. Section 3 presents our approach and the methodology used to compare algorithms and feature engineering techniques. Section 4 describes the data acquisition and cleaning process, while Sect. 5 details the transformations applied to prepare the final datasets. Section 6 introduces the clustering algorithms evaluated and the corresponding metrics, as well as an initial interpretation of the results. Section 7 describes the technical implementation, including frameworks, programming languages, and libraries. Finally, Sect. 8 presents our conclusions and future directions.

2 Related works

Educational data mining (EDM) is the use of data mining applied in the educational field in order to extract meaningful information, relationship, and patterns stored in educational datasets. In the literature of EDM, we can find several works that use not supervised clustering algorithms in order to explore the features, patterns, or factors that make it possible to identify the profiles that make up the target groups of analysis: [4] develops a data-driven clustering model to identify low student engagement during the early stages of the course cycle [5], applies K-means clustering to identify the factors affecting student's academic performance [6], uses hierarchical clustering and K-means to develop different student profiles [7], applies Unique Clustering with Affinity Measure to predict student's behavior in future.

In fact, comparing unsupervised methods across various datasets is a common theme in the literature. For instance [8], evaluates nine well-known clustering methods on 400 artificially generated, normally distributed datasets, underscoring the significance of method comparison in addressing specific problems effectively.

Student dropout is a key topic in Educational Data Mining (EDM) [9], and clustering techniques are commonly used to identify at-risk profiles. Below, we describe several representative studies addressing this issue.

In [10] the authors aim to create a model that identifies students who are at risk of dropping out of high school using information from their first year (i.e., early warning indicators) through machine learning classification algorithms (i.e., Support Vector Machines, Boosting, and Post-LASSO Logit), and then, with the best algorithm-result, apply hierarchical clustering to dropout students to cluster such individuals into different groups and analyze the observable characteristics of each group in order to help design treatments appropriate for each subpopulation.

In [11], the authors predict the outcome of a student (dropout/graduate) in different technical courses. For that, they get the most useful features that may be used to predict a student's performance and explore the features of the different at-risk groups of students through using the K-means clustering algorithm, where "Progress Grade Point Average" result in the most-influencing feature for the detection of different groups of at-risk students.

[12] proposes a new technique, namely self-organizing maps and similarity to an ideal solution (SOM-SIS). This approach utilizes the Technique for Order Preference by Similarity to an Ideal Solution (TOPSIS) [13] to automatically cluster and accurately determine rankings of college academic performance.

The work in [14] aims to uncover hidden patterns in data from the Yemeni university UST (University of Science and Technology) corresponding to academic performance, stu-

dent demographics, and student financial behavior. Based on these data, the authors analyze student dropout by applying clustering (i.e., K-means clustering), association rules (i.e., Apriori algorithm), and Decision Trees. The main conclusions of this study confirm the relationship between dropout and two main features, students' non-payment of fees, and poor results in the first years (first or second). Similarly, in [15] the authors focus on the analysis of students behavior based on the utilization of association rules mining. It aims to identify patterns and relationships within student behavioral data to gain insights into student performance, attendance, access control, and other relevant factors. Based on these patterns, they provide insights to predict student behavior and academic performance, being this specially relevant to identify abnormal situations. Unlike our work, the authors of [15] rely not only on academical information but also on contextual information related to students.

In [16], the authors propose K-means clustering to generate a descriptive model of student dropout in two different contexts: (1) student cases before starting the first year at university and (2) student cases after the first year of study at university. The data used for analysis mainly consist of academic data and basic personal information. This work primarily employs ratios and averages to aggregate all the data. Upon individual analysis of each cluster, the primary findings indicate that males are more prone to dropout than females. Additionally, in the first context, lower Mathematics scores significantly influence the dropout rates. In the second context, lower overall grades predominantly contribute to the dropout rate.

The work in [17] carries out a comparison of different clustering algorithms (K-means, DBSCAN, and BIRCH), as well as a comparison of different types of Feature Selection and Normalization techniques. Then, based on the best results of the different algorithms and feature engineering techniques, they compare classification algorithms that used the results obtained by clustering algorithms to make a more accurate classification. Unlike the work presented here, the authors relied on financial data of students (in addition to academic features) belonging to the lower-income group of Higher Educational Institutions in Malaysia.

In [18] the authors analyze the different reasons that lead students to dropout of higher education in Germany, applying a Hierarchical Clustering and hypothesis testing for it, and using personal and academic data of each student. As the main conclusion, this work indicates that dropout is usually not only due to a single factor but is usually a mixture of factors (parental background, immigration background, study conditions, and personal aspects, etc.). The aim of this work is to support the different higher education institutions to address the different factors they detect in the different clusters and, thus, prevent dropout.

Table 1 presents a comparison of the work mentioned in this section in such aspects as:

- a) Type of data available for analysis;
- b) Clustered algorithms used;
- c) Feature engineering techniques used and compared;
- d) Clustering algorithms to cluster numerical data used and compared;
- e) Clustering algorithms to cluster categorical data used and compared;

As shown in Table 1, numerous studies in the literature employ not supervised algorithms to identify patterns related to student dropout using various types of data, highlighting the scientific interest in this field. However, none of them has provided a comprehensive and exhaustive analysis of the different clustering algorithms found in the literature supported by empirical experiments. Furthermore, there is a lack of research on the impact of feature engineering techniques to transform data into categorical or numerical formats on the application of numerical or categorical clustering algorithms. We explore these kinds of transformations, shedding light on how they influence the performance of the models.

3 Research methodology

This section presents an overview of the research methodology followed to carry out our study. To this purpose, Fig. 1 shows the different steps performed in the analysis, including the techniques and/or algorithms involved in them. Although these steps will be deeply explained throughout the paper (for the sake of replicability of the study), this section introduces a brief summary of each one.

Our approach consists of five main steps from the initial data collection to the final models creation and results comparison tasks. It is worth mentioning that, due to the data quality, the process includes tasks related to data cleaning or Feature Engineering. These tasks could be modified in similar studies depending on the diversity and quality of the data used in such analyses. Feature engineering emerged as a cornerstone of the study, with its emphasis on pre-processing and harmonizing numerical and categorical data effectively preparing the data for the subsequent clustering analysis. Consequently, due to the Feature Engineering potential, changes may significantly impact the obtained results. They will be discussed and compared later on. The main steps of our process are the following:

1. **Fetching data.** Consists of creating a process that connects to the data sources and fetches the data for further processing.
2. **Data cleaning.** Data that may provide noisy information to the models are eliminated, mainly due to being incomplete or not providing correct information.
3. **Feature engineering.** This step deals with several processes on data, such as dealing with redundancy; features significance, correlation, or cardinality; missing values; features creation or elimination; and data fusion. Additionally, different methods such as Binning, Encoding, or Normalization are applied to evaluate the data representation to be provided as input to the different learning models that produce the best results.
4. **Dimensionality reduction.** In this step, dimensionality reduction is performed using the Principal Component Analysis (PCA) algorithm, applied to the feature engineering configuration that yields the best results. The primary objective is to mitigate the noise introduced by the large number of variables generated through one-hot encoding.
5. **Models creation and evaluation.** This part consists of creating the different clustering models. These models will be classified according to the clustering algorithms used (some for numerical variables and others for categorical ones) and the feature engineering techniques applied to provide their input data. Note that we take as input a mixed dataset containing both numerical and categorical features.
6. **Results.** Finally, an evaluation of the models built in the previous step is performed. This evaluation is carried out according to the metrics considered most suitable for the evaluation of clustering algorithms ([19–21]). Also, to validate our approach, we also employ a combination of metrics for labeled data, transcending conventional unsupervised metrics.

The final goal is to explore which combination of algorithm and feature engineering techniques produces the best results.

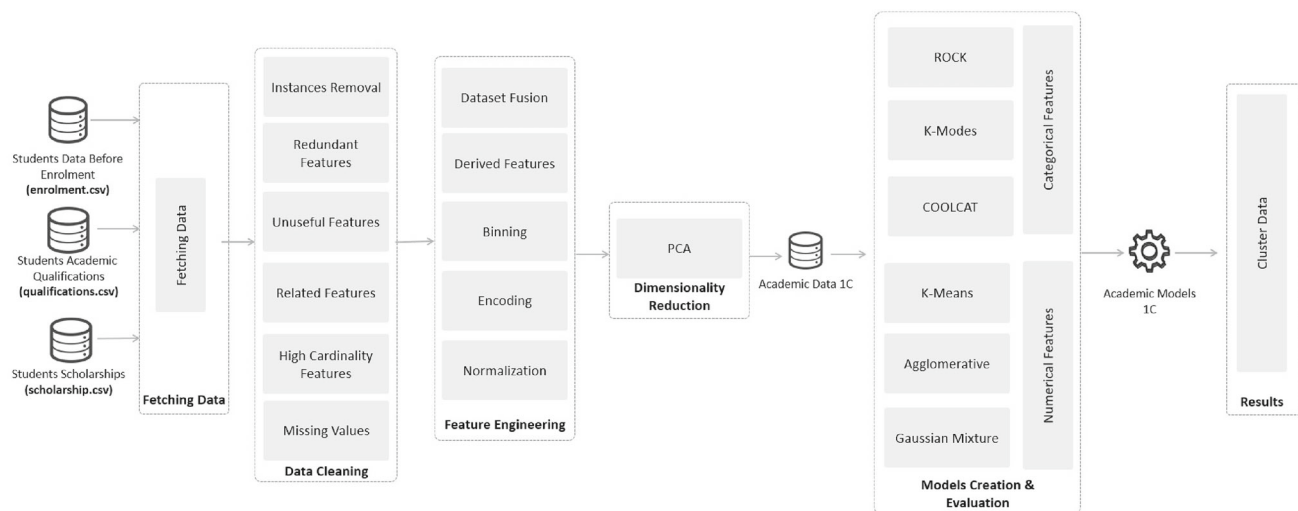
These five steps are described in the subsequent sections. Note that "Data Cleaning" and "Feature Engineering" have been grouped into a single section called "Data Processing" (Sect. 5) to enhance the readability of the paper, as they both involve data processing tasks.

4 Fetching data

The data used in our analysis are mainly provided by a Spanish public university where the analysis is framed within. Concretely, 3 different datasets were provided from an engineering school belonging to this university. The available information pertains to data stored for students who enrolled between 2007 and 2014, with very limited socio-demographic details (limited to age, gender, and municipality

Table 1 Summary of related works

Research work	Type of data	Clustering algorithms	Feature engineering	Clustering numerical data	Clustering categorical data
Sansone et al. (2019) [10]	Academic familiar	Hierarchical	-	-	-
Tamada et al. (2022) [11]	Academic	K-means	-	-	-
Widayanti et al. (2023) [12]	Academic	SOM	-	-	-
Al-shargabi et al. (2010) [14]	Academic financial	K-means	-	-	-
Iam-On et al. (2017) [16]	Academic	K-means	-	-	-
Nafuri et al. (2022) [17]	Academic financial	K-means		K-means	
		DBSCAN	Normalization	DBSCAN	
		BIRCH	Feature selection	BIRCH	-
Behr et al. (2021) [18]	Academic personal	Hierarchical	-	-	-
(Our proposal)	Academic	K-means			
		Hierarchical			
		Gaussian mixture	Normalization		
		ROCK	Binning	K-means	ROCK
		K-modes	Encoding	Hierarchical	K-modes
		COOLCAT	PCA	Gaussian mixture	COOLCAT

**Fig. 1** Overview of the Data Clustering Analysis Process

of residence). Consequently, our study primarily relies on academic data, complemented by a few socio-demographic characteristics.

- Access records (Enrollment).** This data source contains information related to the student accessing data to different degrees (access mark, center of origin, type of access, etc.). It is composed of 5 426 observations and 22 features. It must be taken into account that it contains as many observations as students enrolled in each degree.
- Qualifications.** It provides information about the different evaluations carried out for each subject of the different degrees available for this study (subject, grade,

call, etc.). We rely on 7 undergraduate and 3 master degrees in different engineering fields at the engineering school mentioned above. Its size is 194 569 observations and contains 15 different features. Note that this dataset contains all the evaluations for the different subjects that the students (recorded in the previous dataset) are enrolled in.

- Scholarship.** It contains information regarding the student's scholarship status for each year the student has been studying for a degree (i.e., it indicates whether the student has been awarded a scholarship or not in each year). Its size is 15 760 observations described by 5 fea-

tures. It contains one observation for each year that each student is enrolled in a degree.

The set of available features (name, description, class description, and number) included in these datasets are listed in Table 2. The features that identify each student and thus allow us to relate the information from one dataset to another are "Id" and "Degree Id" (the latter is needed since the former just identifies a student within a degree).

5 Data processing

The application of proper *Feature Engineering* techniques, such as deleting, transforming, merging or splitting features, studying feature correlation, or feature cardinality, among others, is fundamental in machine learning [22][23]. This subsection covers all the aspects related to the application of these techniques, normally encompassed under the term *Feature Engineering* techniques, detailing the reasons that have given rise to their application and how they have been carried out. While we employ a range of data transformations, we particularly focus into the intricacies of handling both numerical and categorical data, specifically examining their impact on the clustering outcomes, but we will work empirically with this in Sect. 6.

Note that the section has been divided into three different subsections according to the purpose of the techniques explained in them: i) data cleaning, used to explain the tasks more related to raw data preprocessing operations; ii) feature engineering, where dataset transformation tasks are described; iii) dimensionality reduction, where we use PCA to reduce the noise introduced by the expanded feature space created through one-hot encoding.

5.1 Data cleaning

All transformations described in this section apply to the three datasets described in Sect. 4 unless stated otherwise.

5.1.1 Instances removal

Raw datasets usually contain instances that may have a detrimental impact on the models ability to learn, leading to models that are incomplete or do not correspond to the reality. In our datasets, there are several cases that, due to few reasons, represent such situations: (1) data that belong to extinct degrees that may jeopardize the results obtained for the new degrees that replaced them; (2) students with incomplete data because of lack of information for their first degree year; (3) students without information about the degree finalization or dropout, since they had not finished the degree when we obtained the dataset.

To palliate the first problem, we removed the data belonging to students enrolled in extinct degrees. For the second one, we removed the data of those students whose did not enroll in the core subjects of the first academic year. Finally, the data of students that we did not have evidences of either dropping out the degree or of finishing it were skipped. After this cleaning process, a total of 3059 out of 5426 students were removed.

Finally, since we were interested in identifying the features that lead to dropping out the degree as soon as possible, for each student we decided to use the academic data corresponding to the first academic year, so that other features (calculated in based on data belonging to higher courses) do not impact on the results obtained.

5.1.2 Redundant features

After analyzing the features in the datasets (Table 2), some were found to exhibit a 1-to-1 relationship, often where a categorical variable and its numerical codification coexist (e.g., Access ID and Access Description, Subaccess ID, and Subaccess Description). In these cases, the numerical feature was removed due to the lack of an ordinal relationship between classes. An exception is Degree ID, retained as part of the primary key for data transformations. Consequently, features such as Access ID, Subaccess ID, Province ID, Municipality ID, Subject ID, and Subject Type ID were excluded.

Additionally, the Call Year feature was removed from the *Enrollment* dataset since its values matched the Enrollment Year for over 99% of instances, retaining only the latter.

5.1.3 Unuseful features

The *Enrollment* dataset includes a feature called dropout which is actually the label feature. The dataset also contains other features such as Closed, Transferred, TransferType, and Blocked. The value of the dropout feature is calculated from these enumerated features, so they can be deleted. However, this assumes that when students transfer to another higher education institution, they still count as dropouts. Unfortunately, we do not have access to information about whether or not these students complete their studies at the new institution. So, in our case, they will be considered as dropping out students (notice that the University is also interested in palliating this situation so that they are also interested in such a situation).

Table 2 Set of features describing the enrollment, qualifications, and scholarship datasets

Dataset	Feature	Description	Class description	# Class
Enrollment	Id	Student ID	Numerical values	1718
	Degree ID	Unique degree identifier	Numerical values	16
	Degree name	Degree name	Categorical: Computer science engineering, civil engineering, telecommunication engineering..	16
	Enrollment year	First academic year of studies	Categorical: 2007-08, 2008-09..	14
	Closed	Indicate whether the record is closed	Binary: Y, N	2
	Transferred	Indicate whether the record is transferred to other institution	Binary: Y, N	2
	Transfertype	Indicates the reason for the transfer of a record	Categorical: internal, external, simultaneous	3
	Blocked	Indicate whether the record is blocked	Binary: Y, N	2
	Call	Call for access	Categorical: june, september..	8
	Call year	Year of the call for access	Categorical: 2007-08, 2008-09, ..	14
	Access ID	Access type unique identifier	Numerical value	-
	Access description	Access type description	Categorical: university entrance examination, Transferred, 25 years of age or older, validated diploma..	11
	SubAccess ID	Subaccess type unique identifier	Numerical value	-
	SubAccess description	subaccess type description	Categorical: LOE, LOGSE, LOMCE..	7
	Marks	University entrance examination	Numerical value between 0 and 14	-
	Origin educational institution	Origin educational institution	Categorical: List of names of the origin high schools	137
	Gender	Student's gender	Categorical: Categorical: H,M	2
	Birth	Date of birth	Date	-
	Province ID	Province ID	Numerical value	-
	Province name	Province name	Categorical: Salamanca, Toledo, Sevilla, Alicante, ..	50
	Municipality ID	Municipality ID	Numerical Value	-
	Municipality name	Municipality name	Categorical: Sevilla, Madrid, Salamanca, Toledo, ..	434
	Dropout	Indicates if the student has dropped out	Binary: Y, N	2
Qualifications	Id	Student ID	Numerical values	1718
	Degree ID	Unique degree identifier	Numerical values	16
	Degree name	Categorical: Degree name	Computer science engineering, civil engineering, telecommunication engineering..	16
	Subject ID	Unique subject identifier	Numerical values	-
	Subject name	Name of the subject	Categorical: physics, linear algebra, data structures and algorithms, calculus, economics and business..	332
	Year	Year	Numerical values between 1 and 4	-

Table 2 continued

Dataset	Feature	Description	Class description	# Class
Scholarship	Semester	Semester of the academic year	Categorical: 1 S, 2 S, Annual	3
	Subject type ID	Subject type unique identifier	Categorical: B, T, O, P, C, E	6
	Subject type Description	Subject type description	Categorical: core, compulsory, optional, internship..	6
	Academic year	Academic year	Categorical: 2007-08, 2008-09..	14
	Call	Call of subject examination	Categorical: june, september, february..	8
	Mark	Mark	Categorical: not taken, fail, compensation, sufficient, very good, outstanding, with honors	7
	Numerical mark	Mark in numerical format	Numerical values from 0 to 10	-
	Attempt	Attempt number	Numerical values from 1 to 6	-
	Id	Student ID	Numerical values	1718
	Academic year	Academic year	Categorical: 2007-08, 2008-09..	14
	Degree ID	Unique degree identifier	Numerical values	16
	Degree name	Degree name	Categorical: computer science engineering, civil engineering, telecommunication engineering..	16
	Scholarship	Indicates if the student has a scholarship	Binary: Y, N	2

5.1.4 Related features

Features were identified as related when a specific value in one feature determined a subset of possible values in another. For instance, in the *Enrollment* dataset, a high correlation exists between *Access Description* and *SubAccess Description*, as the latter is hierarchically dependent on the former. To avoid redundancy and potential negative impacts on the model, we discarded the *SubAccess Description* feature.

5.1.5 High cardinality features

The *Origin Educational Institution* feature from the *Enrollment* dataset contains a high cardinality, as it can take 137 different values, which may lead to high-dimensional spaces or data sparsity [24]. In addition, it has 45.4% null values. Due to these two reasons, the feature *Origin Educational Institution* was discarded.

Similarly, the feature *Municipality* also presents high cardinality problems and its classes are very imbalanced (there are municipalities with up to 500 students, and others with only 4). That means that certain municipalities do not really have sufficient representation in the dataset to extract useful information from them. Therefore, it was decided to

replace this feature with the distance from the study center to the municipality [25]. We considered that this was the best way of using the information provided by municipality (testing whether its distance from the center of studies really influences the results) but trying to avoid the problems related to data imbalance.

5.1.6 Missing values

Missing values are frequent in real-world datasets, and ours is no exception. In our case, we have two datasets with null values:

The *Enrollment* dataset presents null values in the *Marks* feature (38.7% of the cases). In order to address this issue, the null values were substituted with the median value of the corresponding degree (thus, based on the access marks). Median was selected instead of average since it is more robust and less sensitive to outliers.

The *Qualifications* dataset also contains null values in the *Numerical Mark* feature. Additionally, whenever these null values appear in such a feature, the *Mark* feature contains a "Not Taken" value, indicating that the student has not taken the examination. In these cases, we will leave them as null values in the *Qualifications* dataset. For the characteristics calculated in the next subsection, we will use a function that calculates the median while omitting null values.

5.2 Feature engineering

We introduce here the feature engineering techniques applied in order to obtain the best dataset to be later provided to the models.

5.2.1 Dataset fusion

Once the different data cleaning processes were applied to the 3 available data sources, we proceeded to merge the information available in these sources into a single analysis dataset. Thus, this unique dataset contains: (1) the access information of each student (from the *Enrollment* dataset) and (2) the information corresponding to the first year studied by each student (from *Qualifications* dataset), including scholarships information (*Scholarship* dataset).

To this purpose, we considered the different granularity that these datasets present. Therefore, to store the information regarding the first year we had to: (1) obtain for each student whether or not he/she has obtained a scholarship, from the *Scholarship* dataset, (2) group the available information regarding the grades obtained in the first year in different metrics, from the *Qualifications* dataset, (3) add all this information to the accessing data of the *Enrollment* dataset, which is independent of the year.

The metrics that we calculated based on the *Qualifications* dataset to summarize the available information regarding the first year are the following:

- **Success Rate.** This metric measures student performance up to the first course. It is calculated by dividing the number of subjects the student has passed by the total number of subjects the student has enrolled in.
- **Not Shown Rate.** This metric measures the percentage of subjects a student has not taken any examination in up to the first course. It is calculated by dividing the number of subjects in which the student has not taken any examination by the total number of subjects the student has enrolled in.
- **Median Marks.** The median of the marks obtained by a student up to the first course. As previously discussed for the treatment of null values, we use the median as a measure of dispersion and specifically a function that omits the null values. In the case of obtaining a null value as a result, which would indicate that the student has not taken any subject that semester, we will return a value equal to 0 for this metric in order to take it into account in the clustering models applied later on.
- **Pass Median Marks.** The median of the subjects passed in the first course. In addition to the median of the grades, having the median of the grades for the subjects passed and failed separately can also provide valuable information to the model, as claimed in [26].

- **Fail Median Marks.** Grades median for the subjects failed in the first year.
- **Std Deviation Marks.** The standard deviation of grades obtained in first-year subjects. The fact of knowing whether the marks obtained by a student vary greatly from the rest ones or, on the contrary, are similar, also provides information to the model [27].

Therefore, the features of the resulting dataset after merging the 3 available data sources, are: Id, Degree ID, Degree Name, Enrollment Year, Call, Access Description, Marks, Gender, Birth, Distance, Scholarship, Success Rate, Not Shown Rate, Median Marks, Pass Median Marks, Fail Median Marks, Std Deviation Marks and Dropout.

Finally, it must be taken into account that, as we have information available for different years, in this resulting dataset we have the information for students enrolled in different academic years (2007-08, 2008-09,...)

5.2.2 Derived features

Sometimes the available dataset features are only useful if they are properly utilized or derived features are calculated from them. This is the case in our running example of *Birth* feature since, although being a valuable feature, it loses utility when applied in a study along the time, like ours, where different academic years are considered. Therefore, based on the birth date and the year of access, the age at which the student started the degree was derived [27]. Then, *Birth* feature was replaced by the new derived one within the dataset.

5.2.3 Binning

Binning is commonly used in machine learning to transform a numerical feature with continuous values into a categorical feature comprising continuous values into determined ranges.

In this study, the proposed technique is applied to all numerical features, categorizing them into discrete intervals based on the distribution and semantics of each variable:

1. **Grades** are generally grouped into intervals of 1.5 points. However, slight variations exist across different grade-related variables:
 - **Marks** are fully partitioned into 1.5-point intervals.
 - **Median Marks** are divided into two ranges: 1.5–9 and 9–10.
 - **Pass Median Marks** follow a similar scheme, but the first interval spans 0–4.5.

- **Fail Median Marks** are divided into 1.5-point intervals, with the final interval spanning 3–5.
- 2. **Ratios or percentages** are discretized into intervals of 0.25.
- 3. **Age** is grouped into 5-year intervals, up to a final interval of 35–60 years, due to the low representativeness of older age groups in the dataset [28].
- 4. **Distance** is segmented into 45-kilometer intervals up to 250 km, followed by broader intervals from 250 to 500 km, and from 500 km to the maximum recorded distance.
- 5. **Standard deviation** is categorized into three ranges according to the distribution of values in the population: the first range captures the minimum values, the second includes the most common values, and the final range accounts for the maximum values, assuming a roughly normal distribution.

Although alternative discretization schemes and combinations could be explored, the combinatorial complexity renders an exhaustive comparison beyond the scope of this work. Nevertheless, the chosen interval boundaries are not only interpretable but also optimized to maximize intra-interval variance and minimize inter-interval variance. This strategy allows each discretized variable to retain at least 80% of its total variance, in line with standard recommendations [29].

In our case Binning is applied to numerical features because we hypothesize that the ordinal relationships established through the binning process may be more meaningful for the clustering algorithms than the implicit ordinal relationships present in the raw continuous values.

However, being aware that binning may result in some loss of information, we retained the original continuous values. The algorithms were tested using both versions of the dataset: one with binning applied to numerical features and one without. This approach allows us to compare the results and evaluate whether the potential information loss significantly impacts clustering performance (detailed in Sect. 6.3).

5.2.4 Encoding

Many algorithms are unable to process categorical values. This requires the application of processing techniques for them to work with non-numerical features. However, the transformation from categorical features to numerical ones is not exempt from risk and biases that can affect the models performance. There are many encoding techniques that can be employed, the best known being Label Encoding and One-Hot Encoding (both techniques used in works dealing with academic data [27][30]). The former assigns an integer value to each of the feature values, while the latter consists

of creating n new binary features, for each given feature with n classes.

In order to test which encoding technique is better adapted to the analysis we want to apply to this particular dataset, we have tried to apply these two encoding techniques to the different categorical features that make up the analysis dataset for our example. The different algorithms tested and the results obtained are described in detail in Sect. 6.1 and 6.3, respectively.

5.2.5 Normalization

Data normalization is a very common practice in machine learning that consists of transforming different numerical features to a common scale, in order to avoid those features that take large values dominating the learning process even if they are not more important than others in determining the outcome of a model. This technique is necessary when the machine learning technique to be used is sensitive to the presence of different scales in the numerical data, which is common in the vast majority of clustering algorithms designed to work with numerical data.

There are many possible scalers that can be applied. We made use of a simple scaler called *MinMax Scaler*. It scales each feature individually to a given range and we set a [0, 1] range. When using this scaler, the shape of the distribution remains unchanged. If there happened to be *outliers*, they would not be affected. Although this is not always ideal, in our case, there are no significant *outliers* in any of the features, making it a suitable scaler to be applied.

After the Data Cleaning and Feature Engineering process, the analysis dataset used in each of the experiments detailed below has at least the following features: Degree Name, Enrollment Year, Call, Access Description, Marks, Gender, Access Age, Distance, Scholarship, Success Rate, Not Shown Rate, Median Marks, Std Deviation Marks, Pass Median Marks, Fail Median Marks. We discard Id and Degree Id, the features that allow us to identify each student for the experiments and dropout because it is the label we aim to group through clustering algorithms, making it unsuitable for unsupervised learning. After completing this process, the analysis dataset consists of 1773 samples.

5.3 Dimensionality reduction

Dimensionality reduction is the process of transforming high-dimensional data into a lower-dimensional representation that still preserves its essential structure [31].

As noted in the previous section, we encode categorical variables with one-hot encoding. In some of the experiments described later, this expansion brings the feature count up to

88. Reducing the dimensionality of this enlarged space can suppress noise and improve model performance.

Dimensionality reduction methods fall broadly into two families: *linear* (e.g., PCA, LDA, ICA) and *nonlinear* (e.g., UMAP, autoencoders, MDS). In this study we adopt Principal Component Analysis (PCA) [32], the most widely used unsupervised technique by community [31].

To retain a meaningful share of the original information—commonly defined as at least 80% of the total variance [33]—we evaluated several cumulative-variance thresholds (80%, 90%, 95%, and 99%). The best balance between fidelity and parsimony was obtained with a **90% threshold**: the remaining 10% of variance is traded for a potential reduction in noise and improved model performance.

PCA is applied only to the feature engineering configuration that yields the strongest baseline results, aiming to further enhance performance by compressing the final feature set.

6 Models creation and evaluation

After analyzing the literature on clustering algorithms, we observed that many studies have explored the application of clustering techniques for grouping numerical features. However, less effort has been dedicated to the use of such techniques for dealing with categorical features. Additionally, while some works have addressed the application of these algorithms to categorical features [34] [35] [36], their optimization has been largely neglected in the literature

The problem arises, especially when we aim to cluster based on categorical data, but we have dependent features of both types (numerical and categorical). In this context, algorithms designed to work with numerical data do not obtain good results because of the presence of continuous and encoding features (obtained when we pass the categorical features to numerical ones). On the contrary, algorithms designed to work with categorical data do not allow us to exploit the information from numerical features that have been categorized.

For that, we train and evaluate several clustering models as a way to determine which algorithms perform better in terms of the corresponding input dataset and the transformations applied to it. Concretely, the following questions arise: (1) which are the most suitable algorithms for clustering based on a categorical fact with a mixed dataset? (2) what are the appropriate data transformations depending on the algorithms to apply? (3) is it sufficient to convert categorical features into encoding features and apply algorithms dedicated to working with numerical data or, is it better to convert continuous features into categorical features and apply algorithms dedicated to work with categorical data?

This section presents a detailed comparison where different clustering algorithms have been applied to our running example. Note that this example uses a mixed dataset to perform clustering, fixing the number of clusters to 2 (since we are specifically interested in distinguishing between dropout and non-dropout students), and we then compare the resulting clusters with the class labels of the dataset (which are not included as input to the clustering algorithms), following the approach used in other studies [34]. The goal is to evaluate the clustering by examining how well the division aligns with a categorical factor (dropout or non-dropout in our case). Moreover, based on the feature engineering techniques previously explained, different versions of the input dataset have been used so that we may also compare the algorithms in terms of the most suitable preprocessing techniques used. Finally, the results obtained are evaluated in terms of a set of clustering effectiveness metrics so that they may be easily compared.

6.1 Algorithms applied

We divide the clustering algorithms applied in our evaluation into two groups: (1) algorithms designed to work with categorical data, and (2) algorithms designed to work with numerical data. Moreover, as aforementioned, different combinations of Feature Engineering techniques have been applied for each algorithm in order to compare the results obtained and determine which Feature Engineering techniques are most suitable for each algorithm.

6.1.1 Algorithms for categorical data

Since these algorithms only accept input datasets with categorical features, the sole data transformation used for them is the categorization of numerical features into intervals. For this purpose, Binning transformations are applied to these features.

With respect to the algorithms used, they are the following:

1. **ROCK** [35]: The most classical generic categorical data clustering algorithm, based on the use of neighbors and links and using Jaccard's coefficient as a similarity measure.
2. **K-Modes** [37]: It is also a widely used algorithm. It is based on K-means, and it uses the mode as the similarity measure instead of the Euclidean distance.
3. **COOLCAT** [36]: It enables the clustering of categorical data by utilizing data entropy. This approach ensures consistent outcomes regardless of algorithm parameterization, data variability, and dataset size.

6.1.2 Algorithms for numerical data

In the case of algorithms designed to work with numerical data, features may contain encoding categorical values and/or continuous values so that the options for the input data are several:

1. **Use Original numerical data:** This refers to the use of continuous numerical features and encoded categorical features. The encoded values result from applying one of the encoding techniques described in Sect. 5, while the continuous numerical variables (e.g., Marks, Success Rate, Age) retain their original format.
2. **Use Binned numerical data:** This refers to the application of binning to numerical features, dividing their values into intervals. The hypothesis is that the ordinal relationships established through binning may be more meaningful for clustering algorithms than the implicit ordinal relationships in the raw continuous values. After binning, all features are encoded.

Based on these techniques, we had 4 different possible combinations for the input data:

1. Using original numerical data and encoding the ordinal feature *Call* with the Label Encoding technique while encoding the remaining categorical (non-ordinal) features with the One-Hot Encoding technique;
2. Using original numerical data and encoding all the categorical features with the One-Hot Encoding technique;
3. Using binned numerical data applying binning to transform the continuous numerical features into intervals and encoding with the One-Hot Encoding technique;
4. Using binned numerical data by also applying binning and encoding the ordinal features (*Call*, *Marks*, *Access Age*, *Distance*, *Success Rate*, *Not Shown Rate*, *Median Marks*, *Std Deviation Marks*, *Pass Median Marks*, and *Fail Median Marks*) with the Label Encoding technique while encoding the remaining features (non-ordinal) with the One-Hot Encoding technique.

Since these algorithms are sensitive to the presence of numerical values in different ranges, in addition to the above transformations, we apply the data normalization technique mentioned in Sect. 5.2 as an additional final transformation step.

For the best-performing feature engineering configuration among the four listed above, we additionally apply PCA as a dimensionality reduction step, aiming to suppress residual noise and further improve model accuracy.

Regarding the algorithms applied, they are:

1. **K-means** [38]: It is a centroid-based clustering algorithm that starts with an initial suboptimal clustering and progressively enhances it by recalculating the most optimal centroids. The algorithm then assigns data points to the nearest centroids based on the Euclidean distance, effectively relocating them.
2. **Agglomerative** [39]: The method involves a hierarchical clustering algorithm that initially treats each point as a separate group. The algorithm then merges groups together in pairs based on their proximity, using the nearest group criterion, until eventually, all points belong to a single cluster. In this process, the Euclidean distance is utilized as a measure of similarity and Ward as a linkage criterion due to its proven robustness for datasets with mixed variable types [40, 41].
3. **Gaussian mixture** [42]: It utilizes a distribution-based clustering algorithm that defines groups as Gaussian distributions. It assigns individual points to the group with the highest probability of membership, based on the distribution characteristics.

Note that we have applied algorithms with different clustering approaches (centroid-based, hierarchical, and distribution based), enabling the comparison of the results obtained and the selection of the most suitable algorithm for our dataset. Based on the 3 aforementioned algorithms and the 4 possible dataset combinations obtained by applying the data transformations mentioned before, we may obtain 12 results sets based on the application of the algorithms in each case.

6.2 Evaluation metrics

Next, we describe the set of metrics that have been used to compare the results obtained by the different algorithms.

- **Silhouette score** [43]: This metric computes the average Silhouette Coefficient for each point, which is determined by the ratio of the average intra-cluster distance to the average distance to the nearest cluster for each point. It is defined by the following equation:

$$\text{Silhouette Score} = \frac{(b - a)}{\max(a, b)} \quad (1)$$

where

- *a* is the average distance between each point within a cluster (average intracluster distance).
- *b* is the average distance between all clusters (average intercluster distance).

Its possible values range from -1 to +1, where a value of +1 indicates that the particular point is in the correct

cluster, a value close to 0 indicates that the point could belong to another cluster, and -1 indicates that the point is in the wrong cluster.

- **Dunn index [21]:** This metric provides a ratio between the square root of the minimum distance between two clusters and the maximum distance between two points in the same cluster.

It is defined by the following equation:

$$\text{Dunn Index} = \frac{\sqrt[2]{\text{Min Intercluster Distance}}}{\sqrt[2]{\text{Max Intracluster Distance}}} \quad (2)$$

Its range of values is not bounded by a fixed range, but the larger the value, the better the points are considered to be clustered.

- **Davies bouldin score [21]:** It has the positive correlation for the “within-class” case and negative correlation for the “between-class” case. This metric is defined as the average similarity of each cluster to the cluster most similar to it. The similarity is the average distance between two points within a cluster and the distance between one cluster and another. Like the previous metric, it does not have a range of values bounded by a fixed range; however, in this case, the lower the value, the better the clustering performed by the algorithm is considered.
- **Calinski harabasz score [21]:** This metric is defined as the ratio of the sum of the dispersion between clusters to the sum of the dispersion within each cluster. The dispersion between clusters is calculated through the sum of squared distances from the centroid of each cluster to the overall centroid of the data, while the dispersion within each cluster is calculated through the sum of the squared distances of each point to the centroid of its cluster. Its range of values is not bounded by any fixed range, but the higher the value the better the clustering is considered to have been performed.

It is defined by the following equation:

$$\text{Calinski Harabasz} = \frac{\text{BetweenGroup dispersion}}{\text{WithinGroup dispersion}} * \frac{N - k}{k - 1} \quad (3)$$

where

- *BetweenGroup dispersion* measures the weighted sum of squared distances between the centroids of a clusters and the centroid of the whole dataset.
- *WithinGroup dispersion* measures the sum of squared distances between each observation and the centroid of the same cluster.
- N is the total number of points.
- k is the total number of clusters.

- **Impurity [34]:** This metric is commonly employed to assess clustering algorithms that aim to group data based on a categorical feature or variable, utilizing the labels associated with that categorical feature [34, 44, 45]. Once the data have been grouped, they are compared with the class to which each of them belongs. Then, it is considered that groupings with purer clusters, i.e., clusters in which all the points belong to the same class, are preferable. Thus, all those points that do not belong to the majority class represented in each cluster are counted as impurity. This metric obtains a value between 0 and 1 where the lower the result, the lower the impurity in the clusters and therefore, the better the clustering performed.

6.3 Results

Table 3 shows the results obtained in the running example based on the algorithms and metrics presented in previous subsections regarding such aspects as:

- Exp.** Experiment ID where each experiment refers to a specific combination of an algorithm with Feature Engineering techniques;
- Algorithm.** Algorithm Applied;
- Binning.** Use of Binning Techniques;
- Norm..** Use of Normalization Techniques;
- Encoding.** Use of Encoding Techniques (LE stands for Label Encoding and OH for One-Hot Encoding);
- Dim. Red..** Use of Dimensionality Reduction Techniques (PCA);
- Shil.** The value result obtained by the Silhouette Score metric;
- Cal. Harabasz.** The value result obtained by the Calinski Harabasz Score metric;
- David B..** The value result obtained by the Davies Bouldin Score metric;
- Dunn.** The value result obtained by the Dunn Score metric;
- Imp.** The value result obtained by the Impurity metric;
- Clust. 1 0/1** Distribution of the instances in Cluster 1;
- Clust. 2 0/1** Distribution of the instances in Cluster 2

The last two columns show the distribution of instances among clusters according to their class. Instances of students who dropout are associated with class 0, while instances of students who do not dropout are associated with class 1.

Note that Table 3 combines two different evaluation metrics. The first five columns, corresponding to the Silhouette Score, Calinski Harabasz Score, Davies Bouldin Score, Dunn Score, and Impurity metrics, are typical of Unsupervised Learning. They assess the clustering performance without using any external information.

Table 3 Results

Exp	Algorithm	Binning	Norm	Encoding	Dim. Red	Shil	Cal. Harasb	David B	Dunn	Imp	Clust. 1 0/1	Clust. 2 0/1
1	K-means	-	X	LE,OH	-	0.211	507.187	1.851	0.487	0.457	0.43/0.57	0.52/0.48
2	K-means	-	X	OH	-	0.194	448.279	1.968	0.471	0.457	0.43/0.57	0.52/0.48
3	K-means	X	X	LE,OH	-	0.203	480.212	1.902	0.471	0.457	0.43/0.57	0.52/0.48
4	K-means	X	X	OH	-	0.107	217.553	2.836	0.316	0.214	0.22/0.78	0.79/0.21
5	K-means	X	X	OH	PCA	0.116	241.639	2.692	0.296	0.213	0.22/0.78	0.79/0.21
6	Agglomerative	-	X	LE,OH	-	0.167	366.830	2.005	0.442	0.465	0.42/0.58	0.51/0.49
7	Agglomerative	-	X	OH	-	0.152	320.589	2.319	0.442	0.435	0.60/0.40	0.54/0.46
8	Agglomerative	X	X	LE,OH	-	0.162	354.367	2.065	0.346	0.462	0.42/0.58	0.51/0.49
9	Agglomerative	X	X	OH	-	0.099	198.270	2.936	0.408	0.231	0.25/0.75	0.79/0.21
10	Agglomerative	X	X	OH	PCA	0.097	193.397	2.940	0.315	0.258	0.27/0.72	0.77/0.23
11	Gauss. Mixture	-	X	LE,OH	-	0.105	172.834	2.826	0.306	0.477	0.45/0.55	0.49/0.51
12	Gauss. Mixture	-	X	OH	-	0.193	448.279	1.968	0.471	0.457	0.43/0.57	0.52/0.48
13	Gauss. Mixture	X	X	LE,OH	-	0.186	242.455	1.980	0.458	0.477	0.41/0.59	0.49/0.51
14	Gauss. Mixture	X	X	OH	-	0.104	211.048	2.875	0.316	0.221	0.22/0.78	0.78/0.22
15	Gauss. Mixture	X	X	OH	PCA	0.115	237.647	2.704	0.302	0.219	0.22/0.78	0.79/0.21
16	ROCK	X	-	-	-	0.056	31.157	3.100	0.311	0.444	0.15/0.85	0.54/0.46
17	K-Modes	X	-	-	-	0.037	1.109	19.788	0.001	0.361	0.43/0.57	0.87/0.13
18	COOLCAT	X	-	-	-	0.078	106.117	3.962	0.100	0.472	0.50/0.50	0.56/0.44

The Silhouette Score, Calinski Harabasz Score, Davies Bouldin Score, Dunn Score metrics allow us to assess whether the clusters created are compact and whether they are well separated from each other, but do not allow us to assess whether the groups created really correspond to the occurrence of dropout cases. The impurity is a more objective metric that allows clearly seeing whether the corresponding algorithm and applied techniques are able to group the dropout and non-dropout cases, in order to detect the most relevant characteristics of those potential dropout profiles. Thus, this is the metric to optimize and, therefore, the one that will allow us to better define the goodness of the algorithms for our purpose [19, 46, 47].

However, the last two columns, where we compute the belonging of an instance to a cluster, are typical of supervised learning because we are trying to determine how well each model performs on a related task: dropout prediction. These supervised metrics measure how well the model has learned the structure of data related to the dropout task. Notice that we did not use the dropout feature when creating the clusters; it is only utilized for evaluation purposes.

Thus, in addition to understanding how the models perform based on the underlying structure of the data and the identified clusters, we can assess if the defined patterns and structures make sense according to our labeled dataset. This process allows us to gain additional insights into the model's practical applicability with unlabeled data.

Analyzing the last two columns, which contain the results of the ratio of points belonging to each class identified in each cluster, can serve as a good indicator of how well the created model performs in differentiating between the two classes.

Rows 4 and 5 show that 78% of the points identified in cluster 1 belong to class 1 (rightly grouped), while cluster 2 contains nearly the same percentage of instances assigned to class 0. This is the best result, closely followed by the results of row 9. In this best experiment, K-means is used as the algorithm, and processing techniques such as Binning and Normalization are applied along with One-Hot Encoding. The dimensionality reduction also improves (one hundredth in Impurity) the model's result. This example demonstrates how these techniques can significantly boost the results.

With regard to dimensionality reduction, it improves the results in both the K-means algorithm (albeit minimally) and the Gaussian Mixture algorithm. This is not the case with the Agglomerative algorithm, which actually worsens the results.

The model referenced in rows 16 and 17 illustrates how certain models can excel for one class but not for the other. Row 16 demonstrates exceptional performance in cluster 1 by capturing the underlying data structure of dropout, whereas row 17 excels in cluster 2 by capturing the underlying structure of not dropout.

Naturally, the business logic will dictate the choice of model, guided by the cost implications of type 1 errors (predicting dropout when there is none) and type 2 errors (predicting no dropout when there is dropout). Given the favorable trade-off between dropout and not dropout within the clusters, it is reasonable to conclude that the optimal model is the one crafted based on the outcomes from the experimentation in row 5.

Subsequently, we apply PCA [32] also to reduce the dimensionality of the data to two dimensions, facilitating a visual representation of the data distribution through graphs based on distinctive characteristics that inform clustering models (such as Degree Name, Enrollment Year, and others). In Fig. 2, we visualize the dataset in the space of the first two principal components, which together explain 21.6% of the variance (12.9% and 8.7%, respectively), with color distinctions based on the 'Dropout' variable. Conversely, in Fig. 3, the same distribution is colored according to the clusters formed by our best model.

As depicted in Fig. 2, cases of dropout are predominantly concentrated on the right side of the graph (yellow points), while instances of non-dropout are more prevalent on the left side (blue points). Nonetheless, numerous cases of dropout are located on the left side of the graph, where non-dropout cases are more prevalent, and vice versa. Thus, from the data distribution illustrated in Fig. 2, we deduce the presence of dropout cases that share characteristics with non-dropout cases, and vice versa, leading them to occupy similar positions on the graph. This complicates the task of the Model, intended to detect dropout and non-dropout groups, as evidenced in Fig. 3, where it creates a left and a right cluster. While the Model successfully identifies some cases positioned between both groups, it struggles to differentiate certain dropout cases situated amidst a higher concentration of non-dropout cases, and vice versa. This predicament is due to shared characteristics that influence the model formation, resulting in similar placements on the graph. This scenario encompasses approximately 20% of cases within each group, constituting the minority percentage in both clusters formed by Model 5.

Breaking down the results into different methods, techniques, and algorithms, we can analyze them based on the following criteria:

1. Most suitable Feature Engineering transformation techniques;
2. Most suitable algorithm;
3. Combination with the best impurity results obtained;
4. Combinations with the best results for the rest of the metrics.

Feature engineering techniques: We have observed that for algorithms designed to work with numerical features,

Fig. 2 Distribution of data differentiated by Dropout

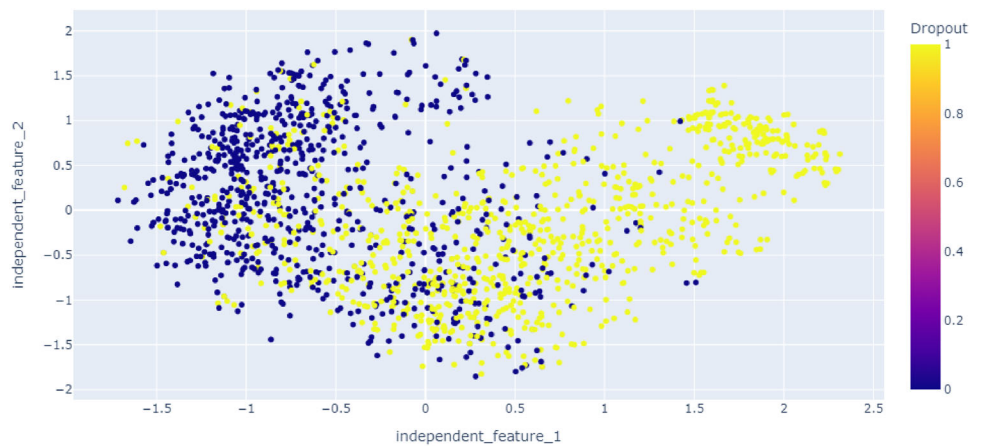
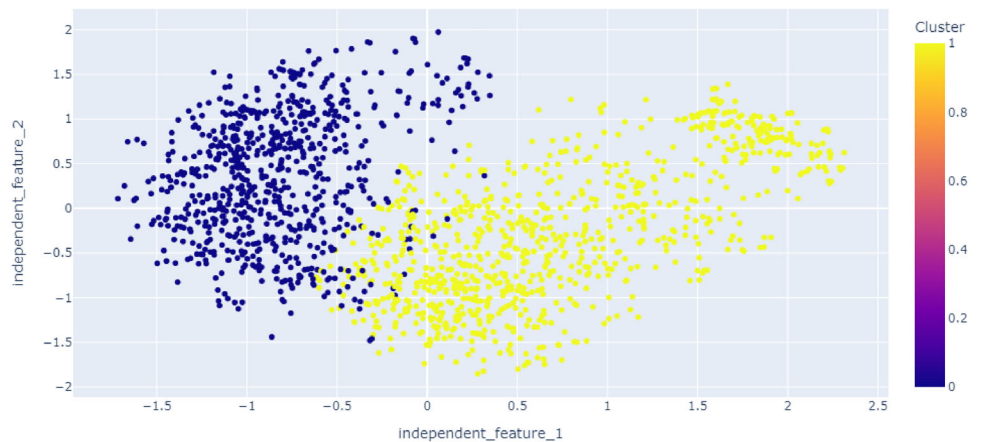


Fig. 3 Distribution of data differentiated by Clusters of model 4



impurity is significantly reduced when applying binning and One-Hot Encoding to all features. This observation is based on the fact that the lowest impurity results within each algorithm designed for numerical features correspond to those where these feature engineering techniques were applied: 5 (K-means), 9 (Agglomerative), and 14 (Gaussian Mixture).

Algorithms: Among the algorithms tailored for numerical data, K-means generally provides the highest reduction in impurity and also delivers the best results in our study. Conversely, among the algorithms developed for categorical data, the K-Modes one exhibits the least impurity in its clusters, achieving over a 11% reduction compared to the Coolcat algorithm and 8% compared to the ROCK algorithm.

Comparing the impurity values achieved by numerical data algorithms versus those designed for categorical data, we find that algorithms meant for numerical data yield superior results. The Binning technique, in particular, stands out for making a significant difference in the outcomes of the experiments. However, the ROCK algorithm also demonstrates a good performance by capturing the underlying data structure of dropout.

Impurity: Impurity has been our key metric for optimization in line with our objectives. It is most effectively

reduced in experiments 4 and 5 (K-means, Binning, Normalization, One-hot Encoding, and, in case of experiment 5, Dimensionality Reduction). This result is accomplished by applying: (1) Binning continuous features into intervals to treat all numerical features as ordinal numerical ones, (2) implementing One-Hot encoding technique followed by the K-means algorithm, and (3) applying PCA as dimensionality reduction technique in case of experiment 5, resulting in an impurity value of 0.214 and 0.213, respectively. This outcome suggests that in the clusters produced through this process, only an average of 21% of the data do not correspond to the majority class within the cluster.

Rest of metrics: Upon examining the experiments in relation to metrics other than impurity, experiment 5 generally yields the most promising outcomes. However, a definitive selection among the experiments according to these metrics is challenging due to the lack of a consistent pattern in their results, i.e., superior performance in one metric does not necessarily translate to superior performance in others. While that experiment might have achieved a better separation and compactness of clusters compared to the one with the lowest impurity value, it fails to form clusters based on dropout;

hence, it does not contribute meaningful insights for this particular study.

6.4 Evaluation

Note that the inconsistent results observed for some metrics may be primarily due to their unsuitability for comparing experiments with different feature spaces. However, we consider all these metrics essential for providing a comprehensive evaluation toolkit. They are especially valuable when comparing similar experiments with varying cluster structures. In this specific case, where labeled data are available, the evaluation approach comprises two components: a combination of clustering evaluation metrics and additional methods that use the labeled data as a baseline for comparison.

To identify the optimal combination of algorithms and preprocessing techniques, we primarily focus on the effective separation of clusters. This process is facilitated by the availability of labeled data, and our findings closely align with the impurity metric results. Impurity serves as a robust measure, independent of how data points are distributed within the feature space, making it particularly suitable for validating clustering results when labeled data are available. This validation approach aligns with the methodologies presented in [34] and [44]. For example, [34] employs impurity (referred to as 'pure' clusters in their work) as a reference metric in experiments on clustering categorical data. Similarly, [44] use both Entropy and Purity (the inverse of impurity) as standard metrics to evaluate and compare different criterion functions for clustering document datasets.

Additionally, to contextualize the results obtained with unsupervised learning, we compared them with those from our previous study using supervised algorithms on the same dataset [3]. The clustering-based approach showed only a slightly lower performance than the best supervised models, despite not using dropout labels during training. This reinforces its applicability as a cost-effective and scalable alternative for early dropout detection in settings where labeled data are limited or unavailable.

Finally, although based on data from a single institution, the proposed methodology is generalizable to other academic contexts, as it relies on standard academic records commonly available across universities. Despite legal and ethical barriers to multi-institutional data access, the study provides transferable insights, and the approach is fully documented to support reproducibility and adaptation.

6.5 Interpretability

Building on the analysis of the best-performing experiment (experiment 5) discussed previously, we conducted a further preliminary analysis aimed at identifying the fac-

tors contributing to cluster formation. It is worth noting that the analysis of feature contributions, along with the potential interventions within the educational setting, lies beyond the primary scope of this study, which is centered on the comparison of algorithm and preprocessing technique combinations. Nonetheless, we conducted this preliminary analysis (intended for further development in future work) to demonstrate the practical applicability of the proposed approach.

To this end, we first examined the importance of the features across the different clusters, followed by an analysis of their individual impact on each student.

6.5.1 Feature importance analysis

Following the approach described in [48], we applied the methodology proposed by [49], which evaluates the importance of each feature in the formation of clusters generated by the K-means algorithm. This is done by analyzing its impact on minimizing the Within-Cluster Sum of Squares (WCSS) metric [50]. Using this methodology, we identified the features with the greatest influence on cluster creation, particularly the cluster predominantly composed of students who dropped out, revealing the following key characteristics:

- Success Rate: Less than 25% of first-year courses passed
- Pass Median Marks equal to or lower than 6
- Marks: University entrance exam score lower than or equal to 6.5
- Fail Median Marks lower than or equal to 1.5
- Not Shown Rate between 50% and 75%, inclusive
- Median Marks lower than or equal to 4.5
- Scholarship false (i.e., the student did not have a scholarship)

However, the features that have the greatest influence on the creation of the cluster predominantly composed of students who do not drop out are:

- Success Rate between 25% and 75%
- Median Marks between 4.5 and 7.5
- Pass Median Marks between 6 and 7.5
- Scholarship true (i.e., the student have a scholarship)

As can be seen, the most decisive factor is the number of subjects passed by the student, where a ratio of less than 25% may be a possible cause of demotivation for the student. On the other hand, beyond the subjects passed and the grades obtained, from a socio-economic point of view, the fact that the student has a scholarship also has an influence.

6.5.2 Individual analysis of cluster representatives

In the previous subsection, we highlighted the features and specific classes within those features that best characterize each cluster. However, these general descriptors can be difficult to interpret at the individual level, especially considering the heterogeneity of each cluster.

The classification of this cluster as primarily associated with dropout is based on the co-occurrence of multiple risk indicators, even though not every student in the cluster necessarily presents all of these features.

However, the transformation of the data using one-hot encoding, normalization, and PCA complicates the direct interpretability of the centroids. An effective approach to improving interpretability is to inspect how close each point is to a cluster centroid and to review its original, untransformed attributes.

By retrieving the actual data points closest to those centroids and interpreting them based on their original, untransformed feature values (i.e., the real categorical and numerical attributes) enable the construction of valid, consistent, and meaningful insights to support decision-making.

For that, we select the 20 points closest to each centroid which provides an optimal balance between interpretability and representativeness. This number corresponds to approximately 1.31% of the instances in each cluster, which is sufficient to capture the dominant characteristics of each group without introducing excessive variability. Choosing fewer than 10 points may lead to anecdotal interpretations that do not generalize well, while selecting more than 30 may start to include edge cases that do not clearly reflect the main structure of the cluster.

Table 4 shows the sample of closest students to cluster centroids. Although the sets of data points closest to the cluster centroids are mostly accurate, 5 cases in the dropout cluster and 3 in the non-dropout cluster do not match the expected label. Rather than removing these cases, we chose to retain them, as they enrich the interpretability of the clusters and support the idea that clustering can reveal underlying student patterns even when the final outcome does not perfectly align with the dominant label.

The cluster identified as "no dropout" is characterized by students with consistent and sufficient academic performance (between 45% and 75% credits passed), median grades (around 6–6.5), and more than 5 passed subjects. Most students exhibit low non-presentation ratios (between 0% and 20%). The majority are scholarship holders (only 3 exceptions).

The cluster associated with student dropout is defined by a notably low cumulative pass rate (often <10%), high absenteeism (frequently >60%), and elevated grade variability, all of which indicate academic instability. Students in this cluster typically have low or borderline median grades (around

5–5.5). A significant portion of them entered university later than usual (ages 27–36), and most do not hold a scholarship (only 3 of them hold it). It can be observed that these findings align with the general conclusions presented in the previous subsection, specially for the features with a higher impact on the clusters formation.

The analysis of the clusters reveals distinct academic and contextual patterns that can be used to guide early intervention strategies. However, as previously noted, this analysis lies beyond the main scope of the study and has been included solely to illustrate the practical applicability of the proposed approach.

7 Materials and methods

The implementation of this project was based on the **MD4DSPRR** framework that we developed in our previous work [51]. This framework is publicly available through the repository `MD4DSPRR utils`.¹

This framework follows a *model-driven approach* that separates data-science projects into two main layers: (1) *conceptual*—definition and design of each pipeline stage—and (2) *operational*—specification of the execution environment. At the conceptual layer, the pipeline is divided into three core stages: **ETL**, **Feature Engineering**, and **Analysis & Modeling**.

The GitHub repository structures data-science projects according to these stages, which can be extended or replaced as needed. In our case, the **Fetching Data** phase was included inside *ETL* (it is essentially an extract–transform–load phase, albeit limited to format standardization). Likewise, **Dimensionality Reduction** was treated as an extra step within *Analysis & Modeling*.

Each stage in the framework is implemented as a class that follows the same three core methods: `load`—to ingest the required datasets; `process`—to perform the data processing operations by calling the relevant functions; and `save`—to persist the resulting datasets. All configuration needed for loading and saving data is provided when each class is instantiated.

By adopting this framework we enforce a standardized project structure and execution environment, thereby enhancing both *reproducibility* and *replicability*.

The framework targets **Python 3.10**, and every pipeline stage is a `.py` file whose parameters are configured via command-line arguments. Consequently, the entire pipeline can be re-run from a single `.yaml` specification.

In our implementation we used **DVC**² to manage that `.yaml` file and reproduce the repository effortlessly. Our

¹ https://github.com/i3uex/apitep_utils

² <https://dvc.org/>

Table 4 Sample of closest students to cluster centroids

Degree name	Marks	Age	Scholar-ship	Success rate	Not rate	shown	Std deviation	Median marks	Pass marks	median marks	Fail marks	median marks	Distance centroid 0	Distance centroid 1	Distance centroid to	Closest Drop-out
Building Eng	6.93	22	1	0.41	0.24	1.92	5.00	5.00	5.00	2.35	1.76	2.75	0	0	0	0
Civil Eng	8.19	23	1	0.44	0.25	2.19	5.69	5.69	7.10	3.20	1.91	3.02	0	1	0	1
Building Eng	6.50	25	1	0.50	0.14	1.93	5.00	5.00	5.89	2.80	1.95	2.89	0	0	0	0
Building Eng	6.50	24	1	0.77	0.08	1.93	6.00	6.00	6.43	2.33	1.95	2.84	0	0	0	0
Civil Eng	6.84	23	1	0.69	0.08	2.13	5.20	5.20	6.00	2.20	1.96	2.75	0	0	0	0
Telecommunication Eng	9.63	21	0	0.50	0.19	1.75	5.50	5.50	6.05	3.00	1.97	2.77	0	0	0	0
Software Eng	8.33	21	1	0.44	0.19	1.45	5.00	5.00	6.00	3.00	2.02	2.91	0	0	0	0
Civil Eng	6.84	23	1	0.47	0.20	2.00	5.15	5.15	5.75	3.10	2.03	3.02	0	1	0	1
Civil Eng	8.11	24	1	0.31	0.13	1.59	3.91	3.91	6.00	3.00	2.05	2.72	0	0	0	0
Building Eng	8.55	21	1	0.28	0.22	1.54	3.60	3.60	5.10	3.30	2.06	2.90	0	0	0	0
Software Eng	10.79	21	0	0.50	0.13	1.80	5.00	5.00	5.90	3.05	2.06	2.91	0	0	0	0
Computer Science Eng	6.54	19	1	0.47	0.20	1.93	5.45	5.45	6.00	3.00	2.07	2.86	0	0	0	0
Civil Eng	7.02	24	0	0.35	0.24	1.84	4.50	4.50	7.25	3.70	2.07	2.82	0	0	0	0
Computer Science Eng	7.31	22	1	0.50	0.19	1.70	5.00	5.00	6.35	3.00	2.08	2.95	0	0	0	0
Civil Eng	6.73	25	1	0.54	0.23	1.58	6.00	6.00	6.06	3.25	2.08	3.05	0	0	0	0
Computer Science Eng	7.90	22	1	0.75	0.00	1.55	5.15	5.15	5.60	3.00	2.08	2.76	0	1	0	1
Telecommunication Eng	6.96	22	1	0.47	0.20	1.62	5.05	5.05	5.50	2.50	2.09	2.92	0	0	0	0
Civil Eng	7.22	22	1	0.35	0.18	1.66	3.65	3.65	6.05	3.00	2.10	2.88	0	0	0	0
Computer Science Eng	6.65	23	1	0.29	0.00	2.01	3.30	3.30	5.80	2.85	2.10	2.72	0	1	0	1
Software Eng	7.03	22	1	0.64	0.14	1.52	5.15	5.15	5.50	3.90	2.10	2.98	0	0	0	0
Civil Eng	5.22	27	0	0.05	0.85	2.11	2.00	2.00	5.90	1.50	2.81	1.92	1	1	1	1
Civil Eng	5.04	21	0	0.11	0.53	1.84	2.20	2.20	5.15	1.30	2.80	1.93	1	1	1	1
Civil Eng	6.61	21	0	0.05	0.55	1.70	2.50	2.50	5.20	1.45	2.68	1.93	1	1	1	1
Telecommunication Eng	10.65	21	0	0.18	0.47	1.98	2.00	2.00	5.60	1.70	2.56	1.98	1	0	0	0
Software Eng	7.92	36	0	0.11	0.61	1.96	2.00	2.00	5.70	1.40	2.70	1.99	1	1	1	1
Building Eng	5.38	24	0	0.05	0.70	1.69	1.20	1.20	5.00	0.80	2.74	1.99	1	1	1	1
Civil Eng	6.84	24	0	0.05	0.79	1.62	2.85	2.85	5.00	2.00	2.62	1.99	1	1	1	1
Civil Eng	6.54	24	0	0.00	0.88	1.12	1.80	1.80	0.00	1.80	2.87	2.00	1	1	1	1
Civil Eng	6.99	22	1	0.18	0.47	1.45	3.00	3.00	5.10	2.25	2.42	2.01	1	1	1	1
Civil Eng	9.19	21	1	0.05	0.79	1.49	2.55	2.55	5.50	2.50	2.45	2.02	1	1	1	1
Civil Eng	6.71	22	0	0.22	0.28	1.94	3.50	3.50	5.55	1.50	2.54	2.04	1	0	0	0
Building Eng	6.50	25	1	0.05	0.63	1.86	2.00	2.00	6.00	1.50	2.53	2.04	1	1	1	1
Computer Science Eng	6.19	31	0	0.22	0.22	1.83	3.00	3.00	5.55	2.50	2.57	2.04	1	0	0	0

Table 4 continued

Degree name	Marks	Age	Scholar-ship	Success rate	Not rate	shown	Std deviation	Median marks	Pass marks	Fail marks	median	Distance centroid 0	Distance centroid 1	Distance centroid to	Closest Drop-out
Civil Eng	6.54	24	0	0.05	0.79	2.15	0.90	5.50	0.80	2.70	2.05	1	0		
Building Eng	5.15	21	0	0.05	0.79	1.13	3.35	5.00	2.80	2.84	2.05	1	1		
Civil Eng	7.37	22	0	0.00	0.80	1.60	1.95	0.00	1.95	2.79	2.06	1	1		
Civil Eng	5.97	25	0	0.18	0.35	1.75	2.70	5.30	2.45	2.74	2.06	1	1		
Civil Eng	5.16	24	0	0.00	0.85	1.35	2.20	0.00	2.20	2.93	2.06	1	1		
Building Eng	6.60	22	0	0.10	0.40	1.60	2.00	5.50	1.80	2.56	2.06	1	1		
Telecommunication Eng	5.57	22	0	0.11	0.56	1.53	3.10	5.00	2.25	2.68	2.06	1	0		

final stage of **Model Creation & Evaluation** was implemented in Jupyter Notebooks and .R scripts (version 4.3.2). The full implementation and configuration details are available to the community on GitHub.³

Data processing relied primarily on the pandas library.⁴ Clustering algorithms were executed through scikit-learn's cluster module.⁵ The only algorithms implemented in R were ROCK and COOLCAT, the former via the cba package,⁶ and the latter via coolcat-r.⁷

8 Conclusion

This study extends our previous research on the early prediction of academic dropout where we used a labeled dataset consisting solely of academic data [3]. This new research shifts its focus toward leveraging unlabeled data, making contributions through this paradigm in the field of student dropout detection without the limitations of having labeled data, which eliminates the need for laborious and costly data labeling efforts.

Primary, the main contribution lies in applying clustering algorithms to analyze the underlying structure of data. Secondly, our study has successfully applied advanced data transformations for the preprocessing and harmonization of numerical and categorical information, acknowledging their distinct characteristics which enhance the robustness of unsupervised algorithms, ensuring that they can effectively extract meaningful patterns from heterogeneous datasets. This part includes the application of Data Preprocessing techniques and Feature Engineering techniques. Finally, we have validated our approach using labeled data, extending beyond traditional unsupervised metrics. This validation has demonstrated the reliability of our results and assessed the generalizability of our methodology, substantiating its potential for practical application in the domain of dropout detection. Additionally, we introduced a preliminary analysis of the features that most influenced the creation of the clusters in the best resulting experiment, providing further insights into the clustering process and its relation to the data structure.

The effectiveness of the proposed approach is validated through a case study involving a real-world dataset of computer science students in higher education. Within this dataset, we formed two clusters that effectively grouped both dropout and non-dropout classes, achieving an accuracy close to 80%.

³ https://github.com/i3uex/education_drop_clustering

⁴ <https://pandas.pydata.org/>

⁵ <https://scikit-learn.org/stable/api/sklearn.cluster.html>

⁶ <https://cran.r-project.org/web/packages/cba/index.html>

⁷ <https://github.com/clbustos/coolcat-r>

For future work, we plan to test clustering algorithms suitable for mixed data, such as k-prototypes [52] and DBSCAN with entropy and probability [53], among others. Additionally, we aim to explore alternative dimensionality reduction techniques, such as UMAP, autoencoders, and MDS, to complement the PCA approach already applied. We also aim to incorporate new data from students in higher academic years and evaluate clustering performance at different academic milestones (e.g., the beginning of each semester). Furthermore, we intend to expand our preliminary analysis and conduct a more in-depth study of the characteristics associated with the identified clusters, with the goal of proposing an early intervention plan tailored to the specific features of students at risk of discontinuing their studies. Finally, we propose developing a follow-up strategy to refine data engineering techniques based on insights gained from the implementation of dropout prevention policies.

Author Contributions All authors helped in conceptualization, methodology, formal analysis and investigation, writing—original draft, writing—review & editing; José M. Conejero, Fernando Sánchez-Figueroa, and Roberto Rodríguez-Echeverría helped in supervision; Fran Melchor and Antonio Jesús Fernández-García curated the data; Fran Melchor, Antonio Jesús Fernández-García, Fernando Sánchez-Figueroa and José M. Conejero validated the study.

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Data Availability Due to privacy limitations, the data used in the study are not publicly available, although they could be available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors declare no Conflict of interest.

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